FINAL EXAMINATION IN GENERAL CHEMISTRY AND PHYSICS FOR ENGINEERS

1. What is the chemical formula of water, the most common liquid on the surface of the Earth?

A. H_2O B. CO_2

C. NaCl D. O_2

Answer: A

2. If $x = 4$, what is the value of $x^2 + 1$ in the expression given in the problem?

A. 9 B. 16

C. 17 D. 25

Answer: C

3. Which quantity is measured in newtons, the SI unit named after the English physicist?

A. mass B. force

C. energy D. power

Answer: B

4. How many protons does a neutral carbon atom have in its nucleus when it is not ionized?

A. 4 B. 6

C. 8 D. 12

Answer: B

5. Which of the following is a vector quantity that has both a magnitude and a direction?

A. speed B. time

C. velocity D. mass

Answer: C

6. What is the acceleration due to gravity near the surface of the Earth, approximately?

A. 9.8 m/s^2 B. 8.9 m/s^2

C. 1.6 m/s^2 D. 98 m/s^2

Answer: A